

— RETICLE · MVP REVIEW

RETICLE

essential

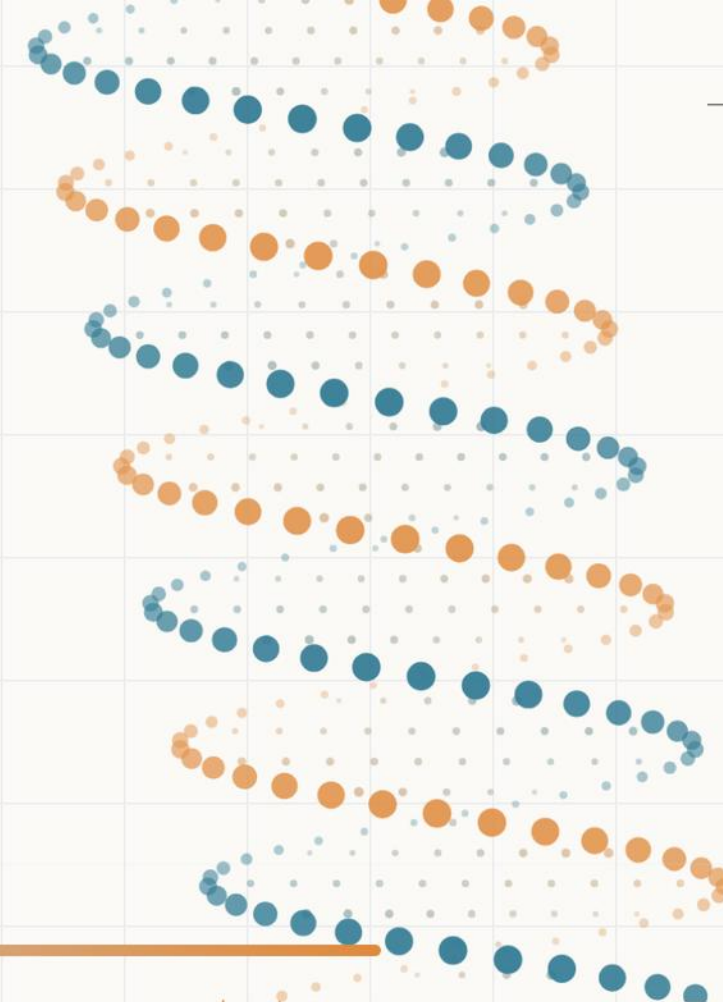
harmonized score

advantageous

One calibrated lens over **2,157** public CRISPR screens — harmonizing scattered functional-genomics data into a single, queryable, AI-ready map of gene function.

RETICLE

MVP presentation



Most of the genome is still *unread.*

up to **80%** of protein-coding
genes have
function that is
**unknown or only
partly understood**
— the “Unknome”

scale two decades after the genome was sequenced, most
genes still have no clear function

clinic patients’ variants stay “**of unknown significance**”
(VUS) because the gene’s job is unknown

signal the functional data to close this gap already exists —
scattered across thousands of CRISPR screens

That experiment already exists — a genome-wide **CRISPR knockout screen** turns off every gene, one at a time. →

*We can switch off every gene
in a single experiment — yet
for most of them, what that
loss actually means is still a
mystery.*

Turn off each gene — one cell at a time.

PERTURB

Knock out

a CRISPR library disables each gene — one per cell, across millions of cells

SELECT

Grow

the pool grows for weeks — on its own, or under a drug or stress

READ OUT

Count

sequence before & after — which knockouts vanished, which took over

dropped out → essential

took over → advantageous

Every screen ends the same way: **one score per gene** — how its loss changed the cell.

Every screen speaks a *different language.*

²¹⁵⁷ public BioGRID ORCS CRISPR screens, run by
hundreds of labs

^{~58} different score types, with sign conventions that
disagree

^{dir} direction is often buried in free-text notes, not the data

^{mix} fitness, stress, and reporter assays all pooled together

*A scientist can read one
screen in a few clicks — but
nothing lets you ask what a
gene does across all of them.*

— 01 — THE GOAL

Put every screen on *one axis*.

Turn 2,157 incomparable screens into a single calibrated coordinate system — **deterministic, reproducible, auditable** — so any gene returns a reliable answer across the whole corpus.



02 — THE SOLUTION · 1 OF 2

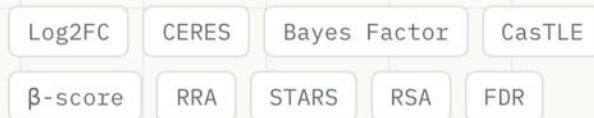
The Logic

How we make 2,157 screens genuinely comparable — before anything reaches the web.

A calibrated score, not a guess.

Harmonized score = each screen's raw effect, aligned to **one shared axis**

~58 RAW SCORE TYPES



SORTED INTO A FEW KINDS

Directional → keep as-is

Directional → flip the sign

Strength only → infer direction

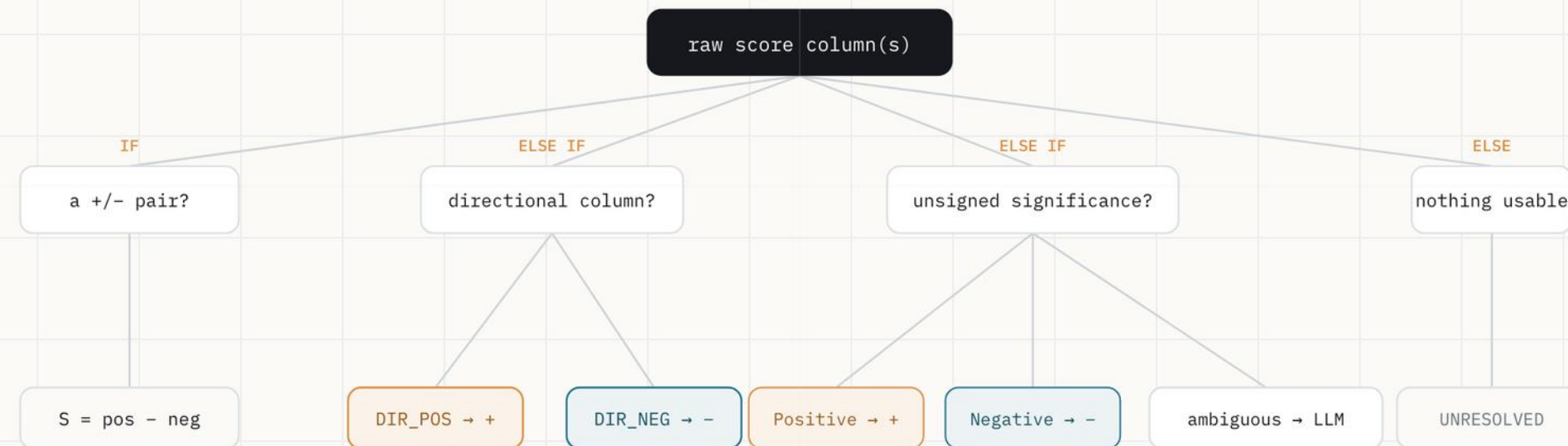
Not a hit score → skip

ONE SHARED AXIS



A fixed list sorts every score type, and a flip aligns knockout vs. activation screens – never guessing from a score's name.

How the score is resolved.



then $\times$ **perturbation** (CRISPRa inverts the sign) $\cdot$ then the **essential-gene anchor** flips the whole screen if core-essential genes land on the + side

The *sign* is the hard part.

A score's *name* doesn't encode its sign convention — so naïve pipelines, and even LLMs, get it **confidently wrong**.

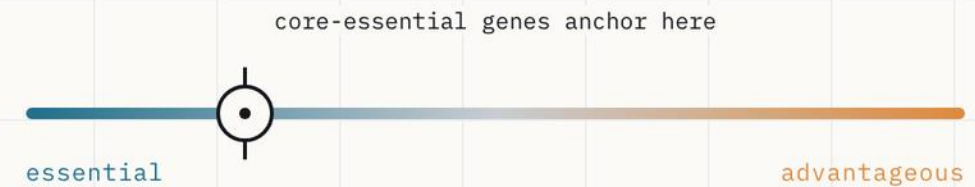
- Same input → **same answer, every time**

fix **Essential-gene anchor.** POLR2A, ribosome, proteasome *must* deplete in a viability screen.

→ If those land on the wrong side, the screen is inverted — flip it. Deterministic & auditable.

llm The LLM resolves only the genuinely ambiguous screens — **freeze** the result.

then we



Watch RETICLE read a real screen.

Uijtewaal ECH (2024) - 5-PMID39681701

Title: CRISPR-StAR enables high-resolution genetic screening in complex in vivo models.

Screen Details:

- Screen Rationale: Cell-essential and growth restrictive genes in established tumors in vivo
- Cell Type: Melanoma Cell Line [BTO:0000849]
- Cell Line: YUMM1.7 [CELLOSaurus:CVCL_JK16]
- Phenotype: Cell Proliferation
- Library: Targeted CRISPRn library in CRISPR-StAR vector [Eiling, 2024] | Type: CRISPRn | Format: Pool | Enzyme: Cas9 | Methodology: Knockout
- Analysis Method: CRISPR-StAR - MAGeCK analysis pipeline | Number of Hits: 139 | Full Dataset Size: 538
- Experimental Setup: Implantation to Mouse Model | Duration: 14 Days | MOI: ~ 0.25
- Significance: Authors have indicated that results with **Score.1 (Log2FC) < -1.5 OR Score.1 (Log2FC) > 1.0** are to be considered significant. These results are indicated by 'Yes' in the HIT column of the table below.

Notes:

- The authors screened for genes whose KO resulted in decreased (negative log2fc) or increased (positive log2fc) proliferation in established tumors from Yumm1.7 450R mouse melanoma cells injected into Rag2-/- mouse models using CRISPR-StAR

[View More Screens from this Publication](#) | [Download Source Supplementary File](#)

CRISPR Screen Results (139 hits)

Showing 1 to 538 of 538 entries

[Download Screen](#)

Score Distribution

Score.1 - Log2FC

Filter these Results...

Filter ✓ Reset ✕

Advanced Filters ▾

Official Symbol	ID Type	Aliases	Organism	Score.1 Log2FC	Score.2 RNAseq	Hit	Rank Order
Atg2b1	gene	2810442122Rik, E130111D10Rik, Pmca1	10090 [M. musculus]	-7.3372	8.003641584	Yes	#1 / 538
Psmb3	gene	AL033320, C10-I1	10090 [M. musculus]	-6.9543	10.09292739	Yes	#2 / 538
Cklr	gene	1700001C14Rik, 1810018M11Rik, A730028G07Rik, C32, CKLF1, CKLF3, CKLF4, CKLF5, Cklp2, Cklr6, HSPC224, UCK-1	10090 [M. musculus]	-6.5744	3.476331557	Yes	#3 / 538
Hspa5	gene	AL022860, AU019543, Bip, D2Wuu141e, D2Wuu17e, Grp78, Hsce70, SEZ-7, Sez7, baffled, mBIP	10090 [M. musculus]	-6.1596	8.361021813	Yes	#4 / 538
Pola1	gene	AW321876, Pola	10090 [M. musculus]	-6.0846	7.679417561	Yes	#5 / 538
Taf1d	gene	2810003M17Rik, 4930553M18Rik, 593042611Rik, Jowd3, TAF1041, TAF141	10090 [M. musculus]	-5.8756	4.159423466	Yes	#6 / 538
Atp5o	gene	ATPO, D12Wuu28e, OSCP	10090 [M. musculus]	-5.8021	7.341273323	Yes	#7 / 538
Tuba1b	gene	Tuba2	10090 [M. musculus]	-5.7465	9.867708284	Yes	#8 / 538
Alp	gene	AA408703, AW476050, Ara9, D19Bwg1412e, Fkbp16, Xap2	10090 [M. musculus]	-5.6286	5.068745594	Yes	#9 / 538
Atg1	gene	ATG1, ATR1, A1010785, AW130871, Atr1, Atrm1, Atr1, ATR1, ATR1, ATR1	10090 [M. musculus]	-5.6286	4.650016831	Yes	#10 / 538

RETICLE READS IT AS

PHENOTYPE

Cell Proliferation → **fitness**

SCORE TYPE

Log2FC → **keep the sign**

METHODOLOGY

Knockout → **no flip**

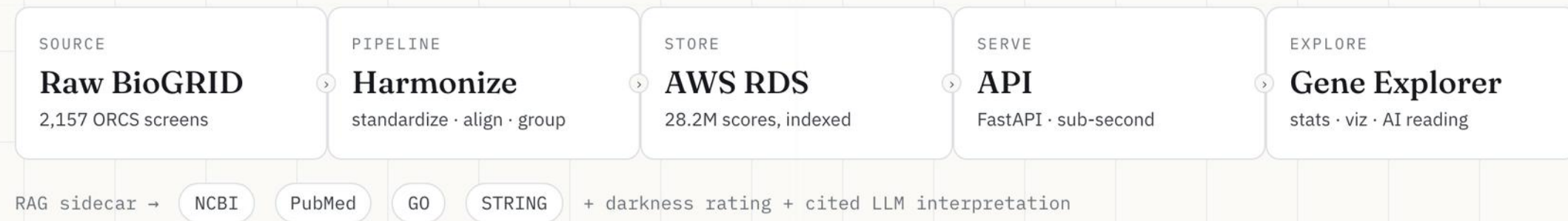
TOP HITS

Psmb3 · Pola1 · Hspa5 → **anchor confirms sign ✓**

BioGRID ORCS · Uijtewaal 2024 · mouse melanoma, in vivo · score: Log2FC

— 02 — THE SOLUTION · 2 OF 2 · THE ARCHITECTURE

From raw screens to a gene you can search.



— 02.2 — THE RAG LAYER

The AI reading is *retrieved*, not invented.

INTERNAL · SCREENING DATA

BioGRID

how the gene **behaves** across 2,157
CRISPR screens — fitness / stress /
reporter, harmonized

EXTERNAL

STRING

who it **interacts** with — top partners +
the evidence behind each link, colored
by their CRISPR behavior

EXTERNAL

NCBI

what it's **known** to do — the curated
gene summary + GO terms

EXTERNAL

PubMed

how **studied** it is — paper count → a
darkness rating; abstracts become
cited evidence

retrieved & ranked → **LLM** → a short interpretation, **cited back to PubMed**

One internal source + three external — every claim traces to a source. The screens stay the source of truth; the model only synthesizes.

— 03 — DEMO

Search a gene. Read its biology.

● Live on AWS RDS — let's open it

Domain-split distribution · STRING
network colored by CRISPR behavior
· darkness rating · cited AI
interpretation.

TP53

Homo sapiens

advantageous

essential

advantageous

SCREENS

1,567

FITNESS

+0.49 n=1008

STRESS

+0.19 n=481

— 03 — WHAT'S LIVE TODAY

28,233,423

harmonized gene × screen scores, loaded
& indexed on AWS RDS

0.2 s

per-gene lookup — down from ~15 s
before indexing

50–90x

faster queries — the index that makes the
explorer usable

2,157

screens harmonized · explorer wired end-
to-end to the cloud database

We're wrapping Aaron's pipeline in a web app researchers can open.

THE STACK

FRONTEND

React app served from **S3** via **CloudFront CDN**

API

FastAPI (Python) container on **AWS ECS**

DATABASE

PostgreSQL on **AWS RDS** — all harmonized data

ETL

WashU **RIS HPC** — where Aaron's pipeline runs

AUTH

WashU **Entra SSO** — scoped to institutional accounts

HOW A QUERY TRAVELS

1

Researcher uploads gene list in the **React** app

2

FastAPI on ECS receives the POST, runs Spearman correlation against RDS

3

Ranked matches, dark gene scores, and graph elements return as JSON

4

React renders results — no page reload, no waiting on a server render

One question, start to finish.

I ran a CRISPR screen and got a hit list — which other published screens look like mine?

You paste in a ranked gene list, pick the organism and assay type, and RETICLE compares it against every harmonized screen in the database — then ranks the matches for you.

• step 3 fires a real POST • results come from RDS

• example gene list is illustrative (28 autophagy genes)



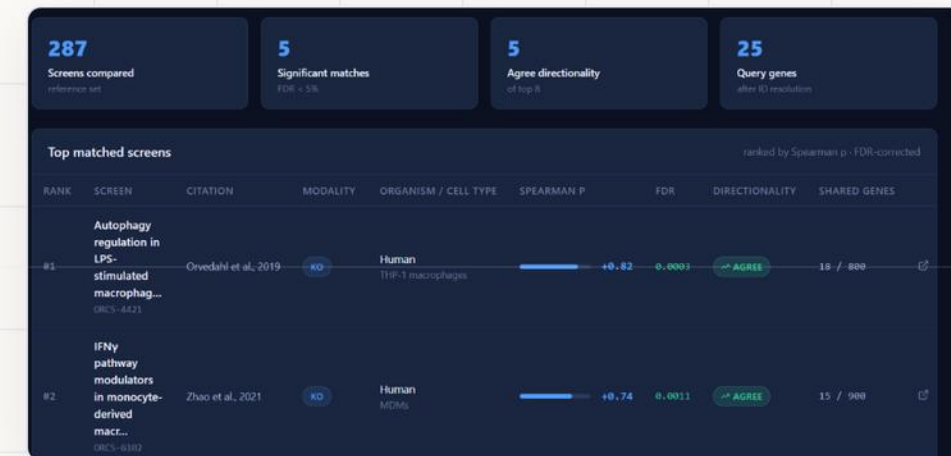
► Demo follows — live walk-through of the app (and the running API if time)

► d2vlieua55yney.cloudfront.net

Under the hood — how the match works.

- 1 Your gene scores are turned into **percentile ranks** — the same step Aaron's pipeline ran on every reference screen, so everything is on one scale
- 2 We compute **Spearman ρ** between your ranked list and each screen, using **only the genes both screens measured**
- 3 Every comparison is **FDR-corrected** and ranked by ρ — strongest matches first

Agree = the same genes mattered in both experiments. **Inverted** = opposite direction (your KO vs. their CRISPRa) — which is biologically informative, not noise.



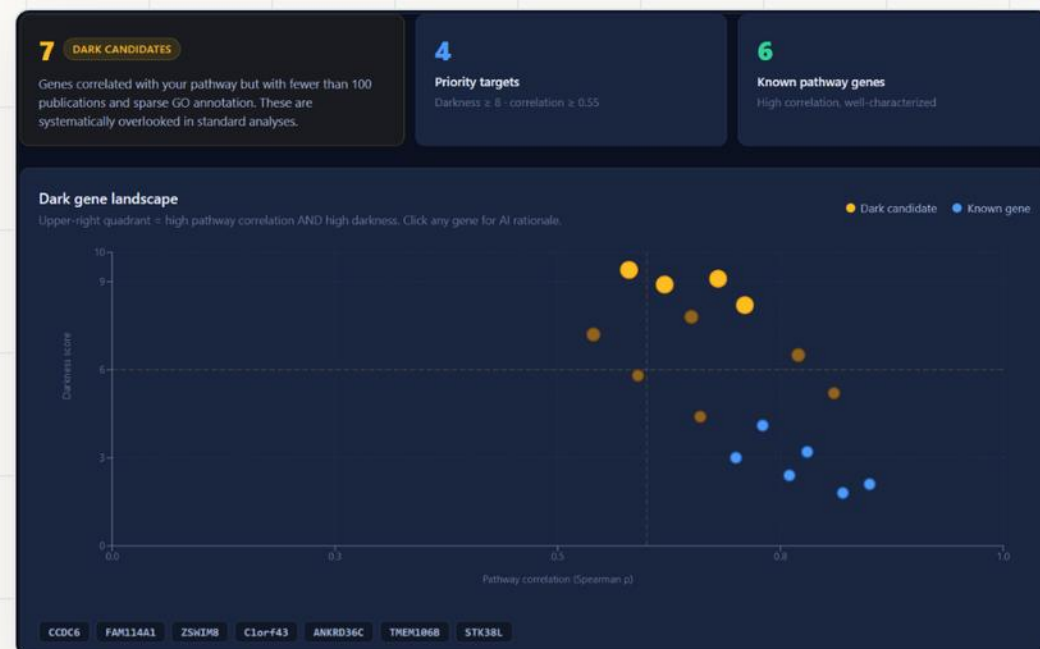
Matched Screens — ranked by Spearman ρ · FDR badges · directionality

The payoff: understudied genes your data points to.

DARK-GENE SCORING

- score** A **darkness score (0–10)** blends publication count, GO-annotation density, and how many of the 2,157 screens ever measured that gene
- plot** We plot each gene by **correlation with your query (x)** against **darkness (y)**
- The upper-right corner is the prize: strongly correlated *and* barely studied

dark scores shown are illustrative • live endpoint being wired



Dark Gene Scatter • correlation (x) vs. darkness score (y)

GRAPH EXPLORER

- nodes** Three kinds of node: **your genes**, the **matched screens**, and the **dark candidates**
- edges** An edge means a gene showed up in a screen; the edge weight is the Spearman ρ
- Click a dark gene and it lights up every screen that flagged it — several independent screens converging is the validation signal

graph wired to API • real nodes & edges from RDS

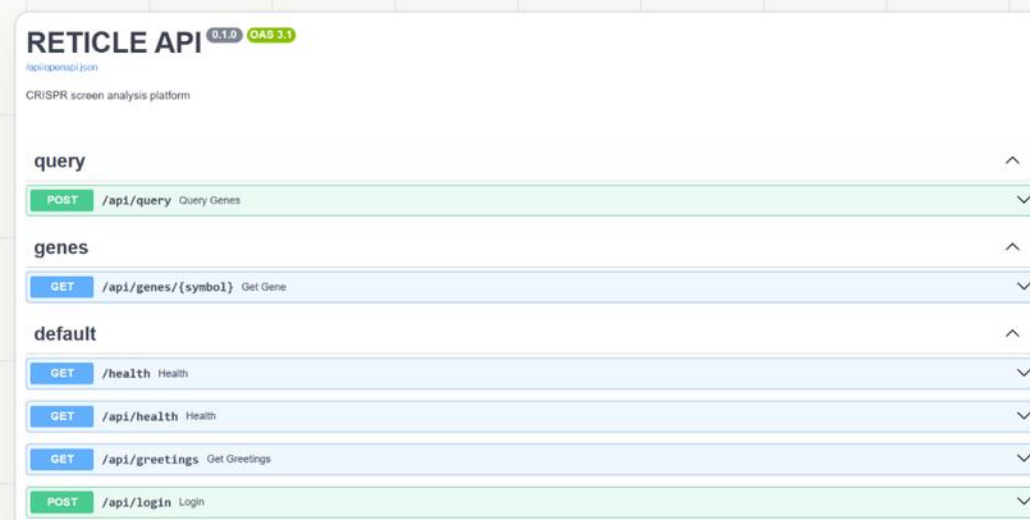


Graph Explorer • screen and gene nodes • Cytoscape.js

Built and running — finishing deployment.

RETURNING REAL DATA

- ✓ **Matched screens** — Spearman ρ , FDR, directionality all live from RDS
- ✓ **Graph explorer** — real nodes and edges from API response
- ✓ **Gene query** — POST /api/query wired end to end
- **API live locally** · wired to RDS

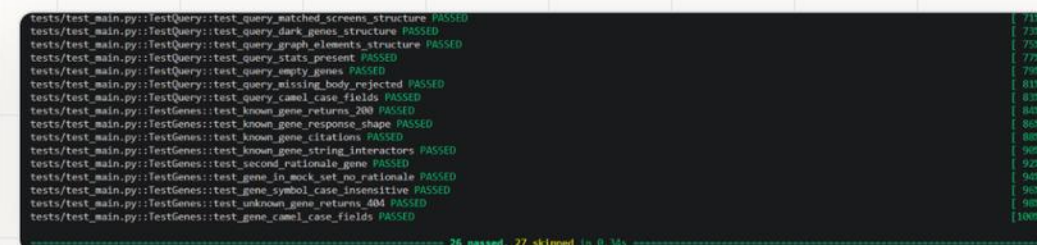


Swagger UI · POST /api/query · GET /api/genes/:symbol

DEMO DATA / IN PROGRESS

- ~ **Example gene list** — 28 autophagy genes, illustrative for demo
- ~ **Dark gene scatter** — UI built · live scores endpoint being wired
- ~ **ECS deployment** — works locally · cloud config in progress
- ~ **MAGeCK .txt upload** — direct file import underway

☛ falls back to demo data if API is unreachable



API response · ranked matches returned from RDS

From working demo to research tool.

01 Dark-gene scoring, end to end

Compute darkness scores from the live database and serve them through a dedicated API endpoint.

02 AI hypothesis generation

A RAG pipeline indexes the source papers so the model can write a cited, mechanistic hypothesis for each top dark-gene candidate.

03 WashU SSO

Wire up Entra for WUSTL login — which then unlocks saved and shareable queries.

04 The full corpus

All 2,157 screens are already harmonized — wire them all up to the API so the webapp queries the complete dataset, not just the working subset.

05 WashU visual identity

Align the UI to WashU brand guidelines — typography, color system, and component style — so it fits naturally in the institutional research ecosystem.